

Mosambee Payment SDK

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TO,

Mosambee Payment SDK user,

We are pleased to send you release in accordance with the details specified below:

This document contains release information for Mosambee Payment SDK, Release Version 1.0.0

Your device must run on Android operating system (OS) 7.0 to 14 (till now).

Version :

- Version Name: 1.0.0

Country:

- India

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1. Required Dependencies for Integration

To ensure the proper functioning of the application, the following dependencies must be included in the **Gradle file**:

1.1. Room Database:

- `androidx.room:room-runtime:2.6.1`
- `net.zetetic:android-database-sqlcipher:4.4.2`

1.2. Coroutines:

- `org.jetbrains.kotlinx:kotlinx-coroutines-android:1.8.1`
- `org.jetbrains.kotlinx:kotlinx-coroutines-jdk8:1.8.1`

1.3. Crypto:

- `androidx.security:security-crypto:1.0.0`

1.4. Network Libraries:

- `com.squareup.retrofit2:retrofit:2.11.0`
- `com.squareup.retrofit2:converter-gson:2.11.0`
- `com.squareup.okhttp3:logging-interceptor:4.11.0`
- `com.squareup.retrofit2:converter-scalars:2.9.0`

2. Required Permissions

Below are the necessary permissions that must be managed by the application implementing the Mosambee Payment SDK.

```
<uses-permission android:name="android.permission.WRITE_SETTINGS"
    tools:ignore="ProtectedPermissions" />

<uses-permission android:name="android.permission.SYSTEM_ALERT_WINDOW" />

<uses-permission android:name="android.permission.CHANGE_NETWORK_STATE" />

<uses-permission android:name="android.permission.BLUETOOTH" />

<uses-permission android:name="android.permission.INTERNET" />
```

```
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />  
<uses-permission android:name="android.permission.BLUETOOTH_SCAN" />  
<uses-permission android:name="android.permission.BLUETOOTH_CONNECT" />  
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>  
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>  
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>  
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
```

In case user is configured for offline transaction below permission is required.

```
<uses-permission android:name="android.permission.MANAGE_EXTERNAL_STORAGE" />
```

3. Additional SDK Integrations

In addition to the dependencies listed above, the merchant is required to implement the following SDKs for payment processing:

3.1. Mosambee Payment SDK:

- This SDK is necessary for integrating the Mosambee payment solution into the application.

3.2. Device SDK and Device Attacher SDK:

- If the merchant wishes to integrate a Mosambee payment device, both the **Device SDK** and the **Device Attacher SDK** must be implemented.
- These SDKs facilitate communication with and management of the payment device.

4. Important Interfaces

4.1. ProcessResult Interface

This interface will allow the application to handle the results of payment commands, including success, failure, and device-specific actions.

Methods:

1. **onCommand(Action action, String command)**

This method is invoked when a command is sent to the payment device. The application can use this to track the command being processed.

Parameters:

Name	Type	Description
action	Action	Represents the action that is being performed (e.g., SALE, UPI).
command	String	The specific command issued to the device.

2. **onSuccess(Action action, JSONObject response, boolean isAckRequired)**

This method is called when the payment process is successful. The response is passed as a `JSONObject`, which contains details about the success.

Parameters:

Name	Type	Description
action	Action	The action that was successfully completed.
response	JSONObject	The response data in JSON format.
isAckrequired	Boolean	Reserved for future purpose.

3. **onFailure(Action action, String reason, String reasonCode, JSONObject jsonObject)**

This method is triggered when the payment process fails. It provides a failure reason, a reason code, and a JSON response for further analysis.

Parameters:

Name	Type	Description
action	Action	The action that was failed to completed.

reason	String	A description of why the failure occurred. (In some cases this may be blank. Please check jsonObject for details)
reasonCode	String	A specific code identifying the type of failure.
responseData	JSONObject	A JSONObject containing additional failure-related details.

4.2.TransactionInteractor Interface

The TransactionInteractor interface defines methods for handling various aspects of a transaction, including tip collection, displaying disclaimers, and verifying a supervisor PIN. The interface provides a structure for interacting with transaction-related data and enables integration with a payment SDK or system that handles these operations.

Key Concepts

- **Tip Collection:** Collects tip amounts from the user during a transaction.
- **Disclaimer Display:** Displays a disclaimer message that the user must acknowledge during the transaction.
- **Supervisor PIN Verification:** Requests a supervisor PIN when elevated permissions are required to approve or modify a transaction. (Reserved For Future Use)

Methods:

1. **tipRequired(String currencySymbol, String currencyCode, Int decimalPlaces, String baseAmount, TransactionDataCollector transactionDataCollector)**

- Parameter Details:

Name	Type	Description
currencySymbol	String	The symbol of the currency in which the tip should be collected (e.g., INR, USD) .
currencyCode	String	The ISO code of the currency (0356 for INR)
decimalPlaces	Integer	The number of decimal places the tip amount should be formatted to, based on the currency's standards.
baseAmount	String	The original transaction amount (before tip), provided as a string to maintain precision.

transactionDataCollector	TransactionDataCollector	An instance of the <code>TransactionDataCollector</code> interface used to pass the collected tip amount back to the payment SDK through <code>onTipCollected()</code> .
--------------------------	--------------------------	--

- Purpose:
This method will be invoked when the system needs to collect a tip from the user. After the tip is collected, the `TransactionDataCollector` will forward the collected amount to the payment SDK for processing.

2. `showDisclaimer(String disclaimer, TransactionDataCollector transactionDataCollector)`

The `showDisclaimer` method is used to display a disclaimer to the user, which they must acknowledge before proceeding with the transaction.

- Parameter Details:

Name	Type	Description
disclaimer	String	The disclaimer message that must be shown to the user.
transactionDataCollector	TransactionDataCollector	An instance of the <code>TransactionDataCollector</code> interface used to notify the payment SDK once the user has acknowledged the disclaimer, typically by calling <code>onDisclaimerResult()</code> .

- Purpose:
This method will display the disclaimer to the user, ensuring that they acknowledge it before proceeding with the transaction. Once acknowledged, the disclaimer information is passed back to the system via the `TransactionDataCollector`.

3. `superVisorPinRequired(TransactionDataCollector transactionDataCollector)`

****This method is reserved for future use.**

5. Connecting Bridge Class

The **MosambeeBridge** class serves as the entry point for interacting with the Mosambee Payment SDK. All SDK methods should be accessed via an instance of this class.

- MosambeeBridge Initialization

To initialize the **MosambeeBridge** class, use the following code snippet:

```
MosambeeBridge mosambeeBridge = new MosambeeBridge(Activity  
activity);
```

- This class should be instantiated in your activity and provides access to all available SDK methods for payment processing and device management.

Note: The `MosambeeBridge` instance must be used for every interaction with the Mosambee SDK. Avoid direct interaction with any other classes in the SDK.

6. Action Processing

The `processActionable()` method is a key function in the Payment SDK used to process any given action related to payments. This method provides a standardized way to handle various actions that can be performed during the payment lifecycle, such as SALE, UPI, or querying transaction statuses.

The method takes two parameters:

Name	Type	Description	Requirement
action	Action	This represents the specific action to be processed (e.g., SALE, UPI, etc.).	Mandatory
bundle	Bundle	A collection of key-value pairs containing the data necessary for executing the given action.	Mandatory

Please refer below sample for `processActionable()` method

```
public void processPayment(){ no usages new *
    Bundle bundle = new Bundle();
    bundle.putString(UserRequestParam.USER, "1234567890");
    bundle.putString(UserRequestParam.PASSWORD, "1321");
    bundle.putString(UserRequestParam.AMOUNT, "1.00");
    bundle.putString(UserRequestParam.BILL_NUMBER, "Test");
    mosambeeBridge.processActionable(Action.SALE, bundle);
}
```

Note: USER and PASSWORD are mandatory parameter for each and every action.

6.1 Actions

In the Payment SDK, every request is processed as an **Action**. To process an Action, you need to refer to the **Action** class present in the Payment SDK.

Each operation corresponds to a specific action that needs to be performed. Below is a list of all available actions in the SDK, which you can use to perform respective operations.

Sr. No.	Action	Description
1.	Action.SALE	This action processes the payment for the goods or services provided to the customer.
2.	Action.VOID	Call when a transaction needs to be cancelled before it is fully completed or settled.
3.	Action.Refund	Call when a customer returns goods or requests a reversal of a completed sale transaction. This action initiates the refund process.
4.	Action.MKE_SALE	Call to initiate a sale transaction for processing with card details when physical card is not available.
5.	Action.MKE_PREAUTH	Call to initiate a pre-authorization for a transaction, reserving funds without completing the transaction with card details when physical card is not available.
6.	Action.MKE_SALE_TIP	Call when processing a tip with a sale with card details when physical card is not available.
7.	Action.MKE_REFUND	Call to initiate a refund for a previously completed transaction. This action processes the reversal of fund with card details when physical card is not available.
8.	Action.EMI	Call when processing an Equated Monthly Installment (EMI) transaction, splitting the payment into monthly installments.
9.	Action.SALETIP	Call when processing a tip with a sale.
10.	Action.PREAUTH	Call when a transaction is being pre-authorized before the actual payment is completed.
11.	Action.SALE_COMPLETE	Call when completing the pre-auth transaction.
12.	Action.MONEY_ADD_ACCOUNT	Call when adding funds to an NCMC card through account typically for virtual wallets or prepaid cards.
13.	Action.MONEY_ADD_CASH	Call when adding funds to an NCMC card through cash typically for virtual wallets or prepaid cards.
14.	Action.BALANCE_UPDATE	Call when updating the balance of NCMC account balance.
15.	Action.NCMC_BALANCE_ENQUIRY	Call when checking the balance of a National Common Mobility Card (NCMC), typically used in public transport systems.
16.	Action.CASH	Call when want to initiate a cash transaction (It will be only for record purpose)
17.	Action.CHEQUE	Call when want to initiate a cheque transaction (It will be only for record purpose)

18.	Action.SETTLEMENT	Call to complete the final settlement of a transaction, transferring funds from the customer's account to the merchant.
19.	Action.RECEIPT	Call to get details of particular transaction
20.	Action.SETTLEMENT_BATCH_DETAILS	Call to get details of open and close batch transactions.
21.	Action.SMS_RECEIPT	Sends a transaction receipt to the customer via SMS. Typically used for mobile payment systems or services with integrated messaging systems.
22.	Action.EMAIL_RECEIPT	Sends the transaction receipt via email to the customer, offering a digital copy of the transaction record.
23.	Action.SETTLEMENT_DATA_PRINTING	Initiates the printing of settlement report.
24.	Action.DETAIL_REPORT_PRINTING	Initiates the printing of detail settlement data
25.	Action.CASH_WITHDRAWAL	Call when a customer withdraws cash from an account. This action processes the withdrawal and updates account balances.
26.	Action.BALANCE_ENQUIRY	Call when checking the balance of an account.
27.	Action.PWCB	Call to perform purchase transaction with cashback amount.
28.	Action.CBWP	Call to perform cash@POS transaction.
29.	Action.TIP	Call when adding a tip to a sale, commonly used in restaurants or service-oriented transactions.
30.	Action.PRINT_RECEIPT	Call when printing a receipt for a completed transaction.
31.	Action.SERIAL_NUMBER	Call when retrieving or recording the serial number of POS terminal.
32.	Action.EMI_SALE	Call when processing a sale that will be paid through Equated Monthly Installments (EMI), dividing the payment into smaller monthly parts.
33.	Action.APP_INIT	Call when initializing application parameter before any transaction processing can occur. (For example primary colour, secondary colour and text colour of application)
34.	Action.CARD_REFUND	Call when a customer returns goods or requests a reversal of a completed sale transaction. This action initiates the refund process with card.
35.	Action.ONLINE_REFUND	ONLINE_ Call when a customer returns goods or requests a reversal of a completed sale transaction. This action initiates the refund process without card.

36.	Action.CARD_NUMBER	Call to get the card details like BIN and SHA of specific card number by swiping the card.
37.	Action.BHARAT_QR	Call when using the Bharat QR code for a transaction. This is used for payments via QR code.
38.	Action.STATIC_QR	Call when wanting to use a static QR code for a payment
39.	Action.UPI_QR	Call when wanting to use a UPI-based QR code for a payment
40.	Action.UPI_VPA	Call when want to process a payment using a Unified Payments Interface (UPI) Virtual Payment Address (VPA).
41.	Action.BHARAT_QR_CHECK_STATUS	Call when checking the status of a Bharat QR-based transaction to verify whether the payment was successful.
42.	Action.UPI_QR_CHECK_STATUS	Call when checking the status of a UPI-based QR transaction to verify whether the payment was processed successfully.
43.	Action.SMS_PAY	Call when wish to initiate a payment via SMS link.
44.	Action.VPA_CHECK_STATUS	Call when checking the status of a payment or transaction initiated with a UPI Virtual Payment Address (VPA).
45.	Action.CARD_TRANSACTION_HISTORY	Call to retrieve a history of card transactions, allowing for tracking of card payments
46.	Action.NON_CARD_TRANSACTION_HISTORY	Call to retrieve a history of transactions not associated with a card, such as UPI, BQR, SMS Pay
47.	Action.LIST_VOID	Call to list all the void eligible transactions, which are not yet settled.
48.	Action.LIST_REFUND	Call to list all the refund eligible transactions
49.	Action.LIST_SALE_COMPLETE	Call to list all the pre-auth transactions, eligible for sale-complete
50.	Action.HISTORY	Call to retrieve a general transaction history, which can include sales, refunds, and voids.

6.2 Card Transaction Processing

The following parameters are required to be passed to the processActionable() method for transaction processing:

Actions: SALE, SALETIP, PREAUTH, CASH_WITHDRAWAL, BALANCE_ENQUIRY, PWCB, CBWP, MONEY_ADD_ACCOUNT, MONEY_ADD_CASH, BALANCE_UPDATE, NCMC_BALANCE_ENQUIRY.

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMOUNT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	Transaction specific description which will be stored against each transaction and can be received in response	Optional
UserRequestParam.CHECK_STATUS_TXN_ID	String	Unique number for each transaction and can be used in check status API from web.	Optional

Notes :

- In case of MPOS device (Devices which need to connect via bluetooth) **UserRequestParam.COMMUNICATION_MODE** parameter need to be pass as "bt".
- In case of SALETIP action TransactionInteractor interface need to be implement. Please refer section 4.2 for more details

6.3 APM Transaction Processing

The following parameters are required to be passed to the processActionable() method for transaction processing:

6.3.1 BHARAT QR

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	Transaction specific description which will be stored against	Optional

		each transaction and can be received in response	
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6.3.2 UPI QR

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMOUNT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	Transaction specific description which will be stored against each transaction and can be received in response	Optional

6.3.3 QR Check Status (Bharat QR /UPI QR)

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.TRANSACTION_AMOUNT	String	Amount for which QR transaction is done. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	Transaction specific description which will be stored against each transaction and can be received in response	Optional
UserRequestParam.REFERENCE_ID	String	Transaction reference id which will be received in every QR generation response as transaction id.	Mandatory
UserRequestParam.TRANSACTION_DATE	String	Date on which QR is generated. Date format is yyyy-MM-dd	Mandatory

6.3.4 Static QR

Name	Type	Description	Requirement
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UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory

6.3.5 UPI VPA

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMOUNT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	Transaction specific description which will be stored against each transaction and can be received in response	Optional
UserRequestParam.BANK_ID	String	VPA address of customer on which collect request need to initiate.	Mandatory

6.3.6 UPI VPA Check Status

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.TRANSACTION_ID	String	The transaction Id of the specific transaction for which check status needs to be done. (This transaction id will be received in UPI VPA response)	Mandatory

6.3.7 SMS Pay

Name	Type	Description	Requirement
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UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMOUNT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	Transaction specific description which will be stored against each transaction and can be received in response	Optional
UserRequestParam.CUSTOMER_MOBILE_NUMBER	String	Mobile number of the customer on which payment link is to be sent.	Mandatory. (If payment link need to sent on mobile number)
UserRequestParam.CUSTOMER_EMAIL_ADDRESS	String	Email address of the customer on which payment link is to be sent.	Mandatory. (If payment link need to sent on email address)

6.3.8 MKE Transactions (Sale /Sale Tip / Pre-auth/ Refund)

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMOUNT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.CARD_NUMBER	String	The credit or debit card number used for making the payment. This is a string of digits that uniquely identifies the user's card for transaction purposes.	Mandatory
UserRequestParam.EXPIRY_YEAR	String	The expiration year of the card (usually a two-digit year, such as 25 for 2025). This is required for validating the card's validity during the transaction.	Mandatory
UserRequestParam.EXPIRY_MONTH	String	The expiration month of the card (usually represented as	Mandatory

		two digits, e.g., "12" for December). This is required for validating the card's validity during the transaction.	
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6.3.9 Cheque

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMOUNT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	The credit or debit card number used for making the payment. This is a string of digits that uniquely identifies the user's card for transaction purposes.	Optional
UserRequestParam.CHEQUE_NUMBER	String	The unique number of the cheque used for the transaction. This is required when a payment is being made via cheque, serving as an identifier for the cheque.	Mandatory
UserRequestParam.CUSTOMER_MOBILE	String	The mobile number of the customer making the payment. This is typically used for communication purposes or sending transaction-related notifications like receipts.	Optional
UserRequestParam.CUSTOMER_NAME	String	The name of the customer making the payment. This parameter is used for identifying the customer and may also be required for transaction documentation or reporting.	Mandatory

6.3.10 Cash

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.AMOUNT	String	Amount for which transaction need to be initiate. (For Example : 1.00, 10.00, 100.00)	Mandatory
UserRequestParam.BILL_NUMBER	String	The credit or debit card number used for making the payment. This is a string of digits that uniquely identifies the user's card for transaction purposes.	Optional
UserRequestParam.CUSTOMER_MOBILE	String	The mobile number of the customer making the payment. This is typically used for communication purposes or sending transaction-related notifications like receipts.	Optional
UserRequestParam.CUSTOMER_NAME	String	The name of the customer making the payment. This parameter is used for identifying the customer and may also be required for transaction documentation or reporting.	Mandatory

6.4 Transaction Reports

6.4.1 VOID LIST || SALE COMPLETE LIST

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory

6.4.2 Card Transaction History || Non-Card Transaction History

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.NUMBER_OF_TRANSACTION	String	Number of transactions to be fetched from server.	Mandatory

Sample Response :

Card Transaction History : - {"result":"success","message":"all card transaction list","responseCode":"00","history":[{"txnId":"2494733932","typeId":"2","amt":"1.00","dateOfTxn":"26/12/2024","timeOfTxn":"12:17:36","rrn":"000000025398","statusId":"2","cardNum":"XXXXXXXXXXXX6438","ctName":"VISA","chName":"SYNERGISTIC FINANCIAL","ssId":"1","invNo":"012666","dccFlow":"1","optinStatus":"0","cCode":"INR"}, {"txnId":"2494731796","typeId":"2","amt":"1.00","dateOfTxn":"26/12/2024","timeOfTxn":"12:17:18","rrn":"000000025399","statusId":"2","cardNum":"XXXXXXXXXXXX6438","ctName":"VISA","chName":"SYNERGISTIC FINANCIAL","ssId":"1","invNo":"012667","dccFlow":"1","optinStatus":"0","cCode":"INR"}, {"txnId":"2494727067","typeId":"1","amt":"1.00","dateOfTxn":"26/12/2024","timeOfTxn":"12:16:37","rrn":"000000025399","statusId":"2","cardNum":"XXXXXXXXXXXX6438","ctName":"VISA","chName":"SYNERGISTIC FINANCIAL","ssId":"1","invNo":"012667","dccFlow":"1","optinStatus":"0","cCode":"INR"}, {"txnId":"2494721559","typeId":"1","amt":"1.00","dateOfTxn":"26/12/2024","timeOfTxn":"12:15:49","rrn":"000000025398","statusId":"2","cardNum":"XXXXXXXXXXXX6438","ctName":"VISA","chName":"SYNERGISTIC FINANCIAL","ssId":"1","invNo":"012666","dccFlow":"1","optinStatus":"0","cCode":"INR"}]}

Non-Card Transaction History : - {"result":"success","message":"all non-card transaction list","responseCode":"00","history":[{"txnId":"2493398763","typeId":"22","amt":"1.02","dateOfTxn":"26/12/2024","timeOfTxn":"06:25:11","rrn":"118560410168","statusId":"2","cardNum":"8779248413@yescred","ctName":"UPI","chName":"NA","ssId":"4","invNo":"000484","dccFlow":"1","optinStatus":"0","cCode":"INR"}, {"txnId":"2479611357","typeId":"22","amt":"1.00","dateOfTxn":"24/12/2024","timeOfTxn":"02:54:11","rrn":"118361522006","statusId":"2","cardNum":"8779377651@yescred","ctName":"UPI","chName":"NA","ssId":"4","invNo":"000483","dccFlow":"1","optinStatus":"0","cCode":"INR"}, {"txnId":"2445263730","typeId":"37","amt":"1.01","dateOfTxn":"19/12/2024","timeOfTxn":"04:17:41","rrn":"117863668978","statusId":"2","cardNum":"subramanianganesh7@okaxis","ctName":"UPI","chName":"NA","ssId":"4","dccFlow":"1","optinStatus":"0","cCode":"INR"}, {"txnId":"2445263719","typeId":"37","amt":"1.01","dateOfTxn":"19/12/2024","timeOfTxn":"04:17:36","rrn":"117863643718","statusId":"2","cardNum":"subramanianganesh7@okaxis","ctName":"UPI","chName":"NA","ssId":"4","dccFlow":"1","optinStatus":"0","cCode":"INR"}, {"txnId":"2445263608","typeId":"22","amt":"1.01","dateOfTxn":"19/12/2024","timeOfTxn":"04:16:39","rrn":"117863641583","s"}]}

```
tatusId":"2","cardNum":"subramanianganesh7@okaxis","ctName":"UPI","chName":"NA","ssId":"4",
"dccFlow":"1","optinStatus":"0","cCode":"INR"]}]}
```

****Please refer section 7 for more details**

6.4.3 Transaction Receipt

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.TRANSACTION_ID	String	The transaction Id of the specific transaction for which data needs to be retrieved.	Mandatory

6.4.4 Settlement Batch Details

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.BATCH_NUMBER	String	Pass "0" for open batch details and "2" for closed batch details.	Mandatory

6.5 Other Transactions

6.5.1 Void Transaction

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.TRANSACTION_ID	String	The transaction Id of the specific transaction which need to be voided.	Mandatory

6.5.2 Sale Complete Transaction

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.TRANSACTION_ID	String	The transaction Id of the specific transaction against which sale complete need to be initiated.	Mandatory
UserRequestParam.AMOUNT	String	Transaction amount of which sale complete needs to be performed.	Mandatory

6.6 Others

6.6.1 Serial Number

To get the device serial number need to pass empty bundle to SDK

6.6.2 Send SMS Receipt

Name	Type	Description	Requirement
UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.TRANSACTION_ID	String	The transaction Id of the specific transaction which need to be voided.	Mandatory
UserRequestParam.MESSAGE_TO	String	This parameter is used to specify the mobile number of the customer to whom the digital receipt link will be sent.	Mandatory

6.6.3 Send Email Receipt

Name	Type	Description	Requirement
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UserRequestParam.USER	String	User name provided by Mosambee which will be unique for each merchant	Mandatory
UserRequestParam.PASSWORD	String	Password provided by Mosambee	Mandatory
UserRequestParam.TRANSACTION_ID	String	The transaction Id of the specific transaction which need to be voided.	Mandatory
UserRequestParam.EMAIL_TO	String	This parameter is used to specify the email id of the customer to whom the digital receipt link will be sent.	Mandatory

6.6.4 Print Receipt

Name	Type	Description	Requirement
UserRequestParam.PRINT_RECEIPT_DATA	String	Receipt data(JSONObject as string) which will be received in payment response or fetched from history.	Mandatory
UserRequestParam.IS_MERCHANT_COPY	Boolean	Whether to print "Merchant Copy" on receipt. true – "Merchant Copy" false – "Customer Copy"	Mandatory
UserRequestParam.IS_DUPLICATE	Boolean	Whether to print "Duplicate" on receipt	Mandatory

** Print Receipt will require entire response data.

7. History/List API Parameters

Sr. No.	Parameter Name	Description
1.	txnId	Transaction id of transaction.
2.	typeId	Transaction Type Id (Refer section 8 for more details)
3.	amt	Transaction amount
4.	dateOfTxn	Date of transaction.
5.	timeOfTxn	Time of transaction.
6.	rrn	Retrieval reference number of transaction. (unique identifier associated with a specific transaction)

7.	statusId	Status of transaction. (Refer section 9 for more details)
8.	cardNum	Card number in case of card transaction. UPI id in case of UPI transaction.
9.	ctName	Card type name (For example: VISA, MASTER,... Incase of card transaction)
10.	chName	Card holder name
11.	ssId	Settlement status id (Refer section 10 for more details)
12.	invNo	Invoice number of transaction
13.	dccFlow	Whether the transaction is done with DCC flow or not
14.	optInStatus	Opt in status of the transaction (Use only in case of dcc)
15.	cCode	Currency code of transaction

8. Transaction Types

Transaction Type Id	Transaction Name
1	SALE
2	VOID
3	SETTLEMENT
4	REFUND
7	TIP
8	PREAUTH
9	PREAUTH COMPLETE
10	CASH
11	CHEQUE
13	PWCB (Purchase Without Cash Back)
14	CBWP (Cash Back Without Purchase / Cash@POS)
19	SMS PAY / PAY BY LINK
22	UPI

24	BHARAT QR
27	SALE_TIP
29	BALANCE ENQUIRY
33	MONEY ADD ACCOUNT
34	MONEY ADD CASH
35	BALANCE UPDATE
36	CASH WITHDRAWAL

9. Transaction Status

Transaction Status Id	Transaction Status
1	INPROCESS
2	APPROVED
3	DECLINED
4	SETTLED
5	REVERSED
6	SETTLEMENT FAILED
7	SIGNATURE PENDING (Transaction is successful but signature not being captured by merchant.)
8	REVERSED (Decline transaction is reversed.)

10. Settlement Status

Transaction Status Id	Transaction Status
1	UNSETTLED
2	SETTLED
3	FILE BASED SETTLEMENT
4	NOT APPLICABLE FOR SETTLEMENT

11. Error Codes

11.1 Terminal Level Errors

Error Code	Error Message	Description
EE01	Transaction Timeout	Transaction initiated but no card presented for transaction.
EE02	Pinpad Timeout.	Transaction initiated and card presented for transaction but not entered pin for 45 secs.
EE03	Application Blocked.	For specific test case where application present in card is blocked. (Error return by terminal)
EE04	Card Blocked.	For specific test case where card is blocked at EMV level. (Error return by terminal)
EE05	Card Blocked.	For specific test case where card is blocked at EMV level. (Error return by terminal)
EE06	Card Reading Failed. Please Try Again.	Due to any reason if card is not properly read.
EE07	Unsupported Card. Please Try With Another Card.	If chip card is not properly read. And the terminal do not support the fall-back transaction.
EE08	Transaction Cancelled By User.	If transaction is initiated and purposely cancelled by user.
EE09	Transaction Cancelled.	If pin-pad is cancelled by user.
EE10	Expired Card.	If expired card is identified by terminal level. (Please note not every card is declined by expired card error at terminal level.)

EE11	Transaction Terminated.	Transaction terminated by POS device.
EE12	Keys Not Present. Require Key Injection.	If the POS device is not equipped with the DUKPT keys, please contact Mosambee for key injection into the terminal.
EE13	Contactless Card Reading Failed. Please Try Again.	If the contactless card reading fails, it could be due to the card not being properly tapped or an issue with the card itself.
BT_ERROR_01	BT_ERR_01- Please enable bluetooth	When a user attempts to initiate a transaction with the MPOS device but the Bluetooth on the mobile phone, through which the MPOS is connected, is turned off.
BT_ERROR_02	BT_ERR_02- Kindly pair mosambee supported bluetooth device for proceed transaction.	When a user attempts to initiate a transaction with the MPOS device but the device is not connected with mobile phone.
DEVICE_ERROR_01	DEVICE_ERR_01- Device not supported	When user is initiating transaction on unknown device which is not supported by Mosambee Payment SDK

11.2 Network Level Errors

Error Code	Error Message	Description
N_ERR - 500	Unable To Process Your Request	If service face any internal server error.
N_ERR - 503	Unable To Process Your Request	The requested service is not available at the moment
N_ERR - 400	Unable To Process Your Request	The server cannot process the request due to a client error.
N_ERR - 401	Unable To Process Your Request – Unauthorised	The request requires user authentication. The client must provide credentials to access the resource.
N_ERR - 403	Unable To Process Your Request	The server understands the request but refuses to authorize it.
N_ERR - 404	Unable To Process Your Request	The requested resource could not be found on the server
	Request timed out. Please check your internet connection and try again.	When server request timeout due network issue or some other issue at service level. (Socket Timeout)
	A network error occurred. Please check your internet connection and try again.	When face any socket exception.
	An error occurred while processing your request. Please try again later.	When request could not be processed due to IO exception.
	An unexpected error occurred. Please try again later.	An unhandled network error.

S_ERR - 00	Unable To Process Your Request	When received blank response from server.
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